COURS DE MECANIQUE
Milieux curvilignes

Statique des milieux curvilignes

10 septembre 2026

Ecole Nationale des Ponts et Chaussées

Structure générale du cours

module 1 :
- Mécanique des milieux curvilignes (obligatoire)
- Mécanique des milieux continus tridimensionnels (obligatoire)

module 2 : Approches variationnelles et numériques (électif)

module 3 : Introduction à la Mécanique non linéaire (électif)

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Sécurité

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Sécurité

Confort

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Sécurité

Confort

Calcul des déplacements

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Sécurité

Confort

Calcul des déplacements

↓

déformée

Objectifs du cours

Calcul et dimensionnement des structures et ouvrages

Sécurité

Calcul des efforts intérieurs

Confort

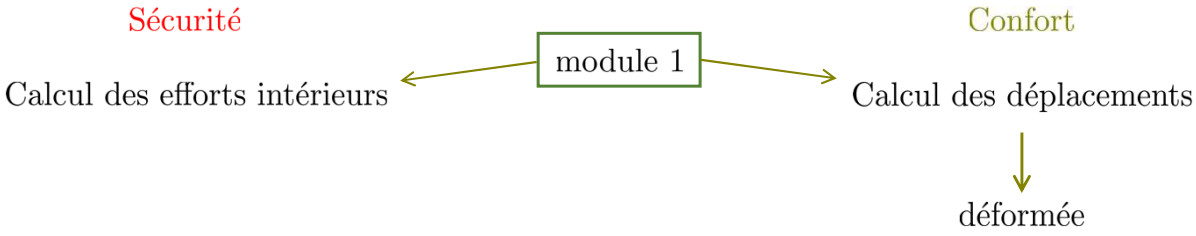
Calcul des déplacements

↓

déformée

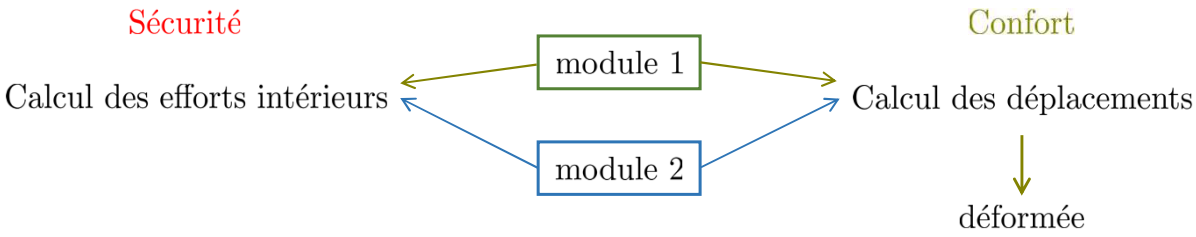
Objectifs du cours

Calcul et dimensionnement des structures et ouvrages



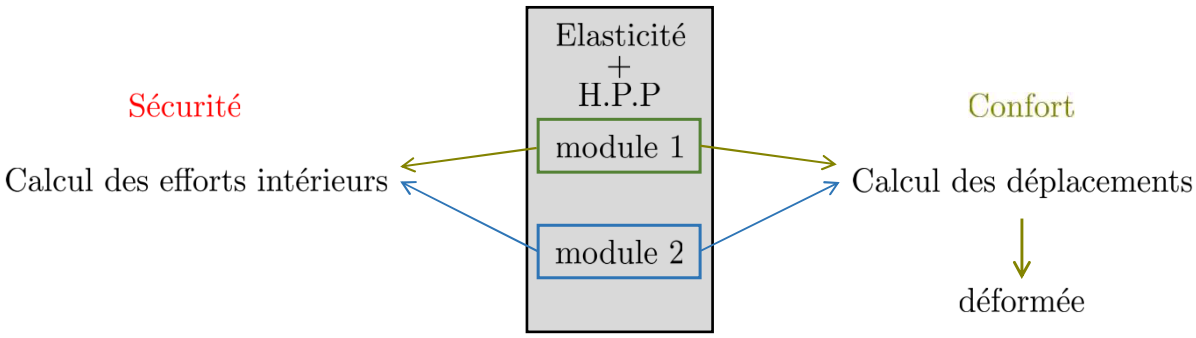
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Calcul et dimensionnement des structures et ouvrages



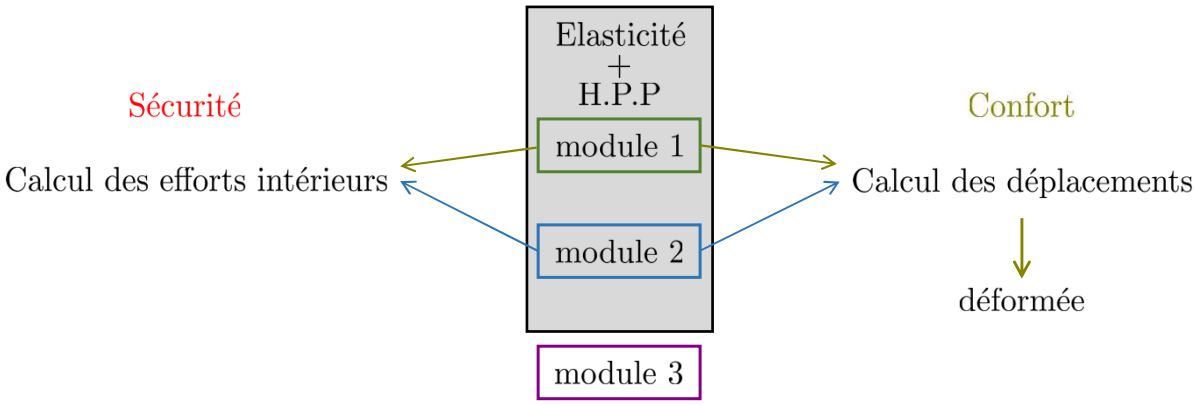
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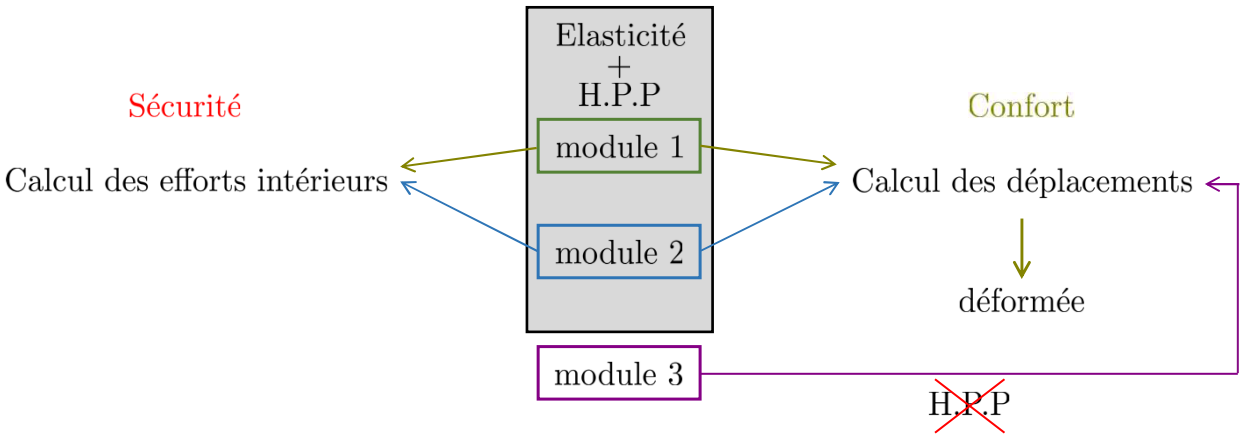
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Calcul et dimensionnement des structures et ouvrages



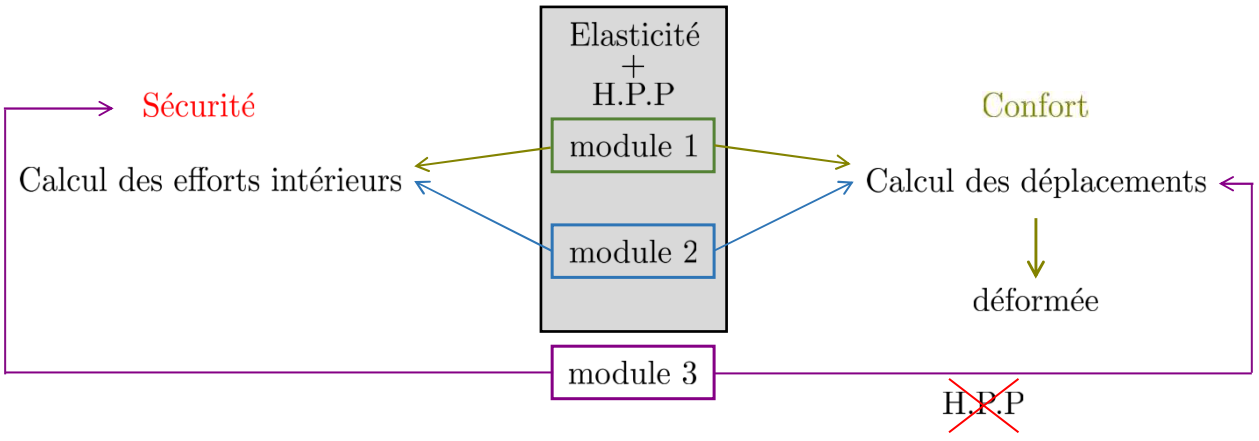
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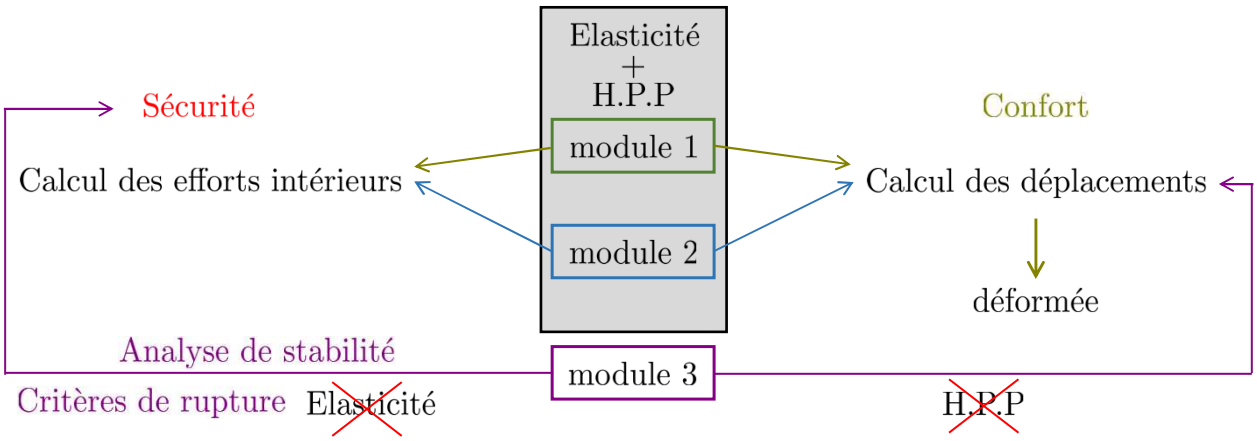
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Objectifs du cours

Calcul et dimensionnement des structures et ouvrages



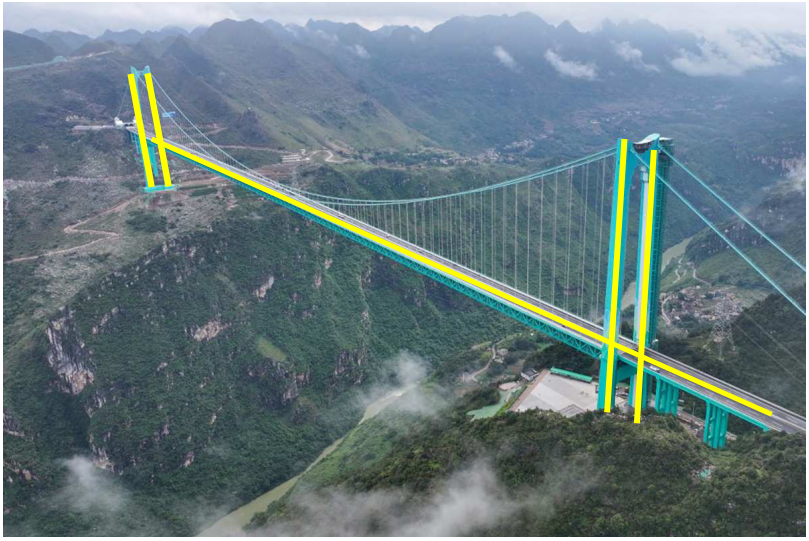
Milieu curviligne : modélisation géométrique 1D d'un solide 3D élancé



Milieu curviligne : modélisation géométrique 1D d'un solide 3D élancé

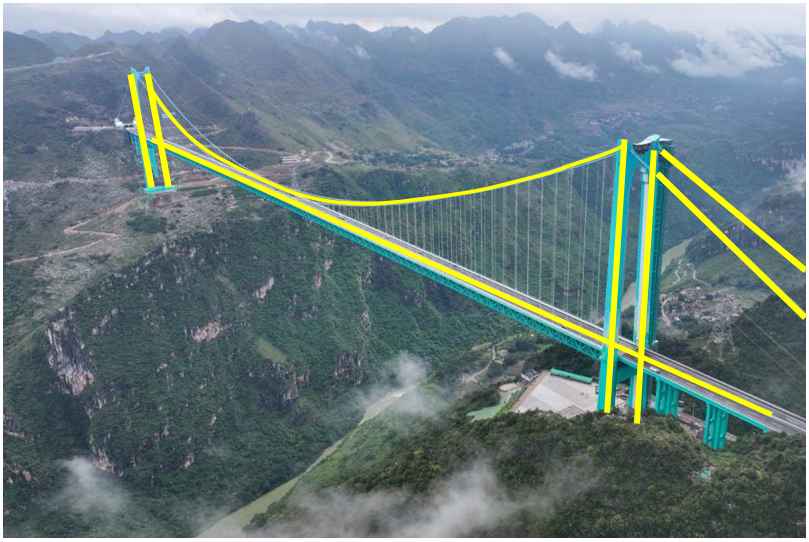


Milieu curviligne : modélisation géométrique 1D d'un solide 3D élancé

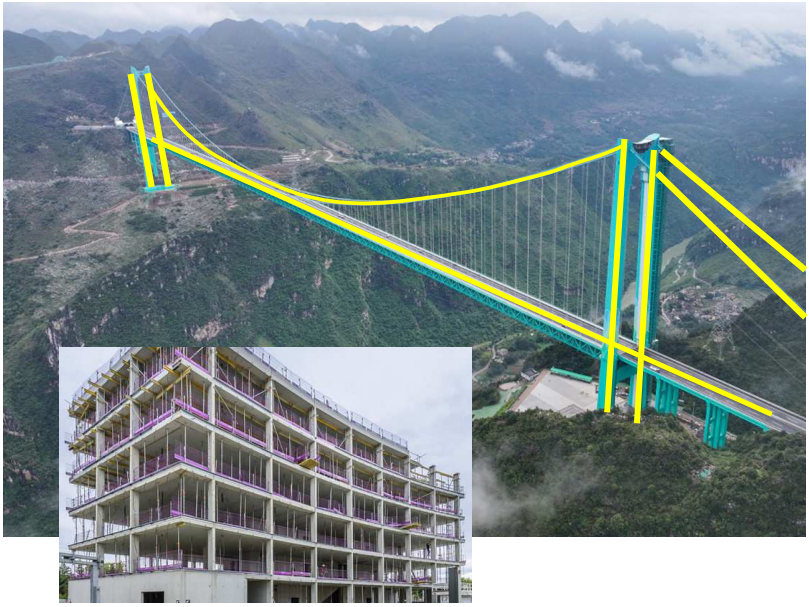


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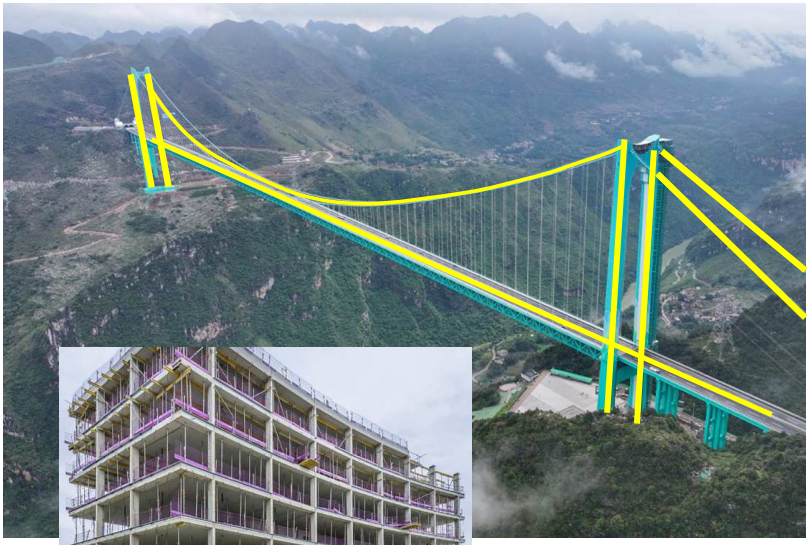
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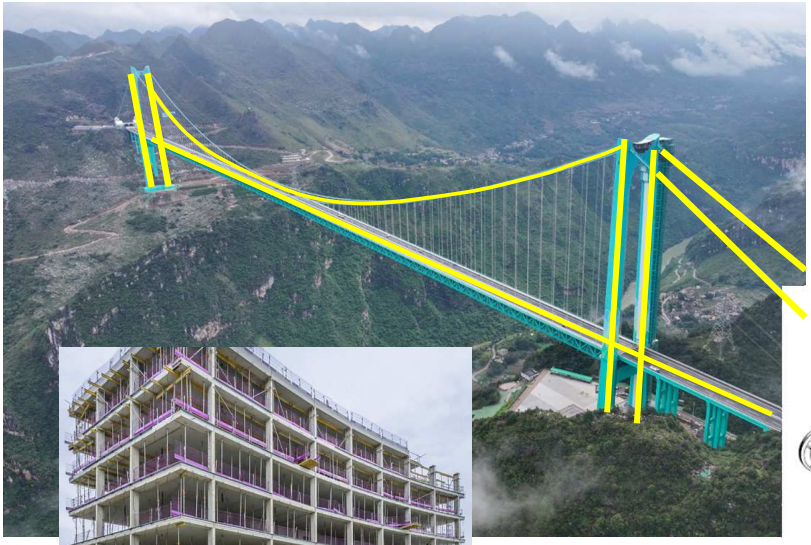
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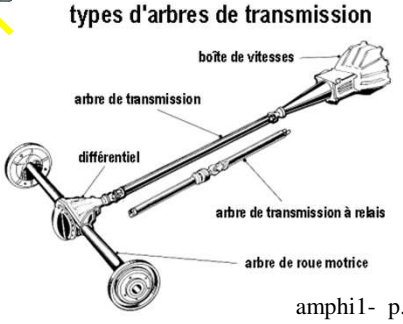


Milieu curviligne : modélisation géométrique 1D d'un solide 3D élancé









Plan du module 1 : mécanique des milieux curvilignes

séance 1 : Statique des milieux curvilignes (CM + TD)

séance 2 : Statique (suite), cinématique et TPV (CM + TD)

séance 3 : Déformations, comportement élastique et résolution directe (CM + TD)

séance 4 : Méthodes de résolution énergétiques (CM + TD)

séance 5 : Exercices de synthèse (TD)

2 séances de TPA

contrôle des connaissances

Outils pédagogiques

- page Educnet
- documents projetés en amphi
- TD : sujets et corrigés
- polycopié synthétique

Equipe enseignante

CM : Ghazi HASSEN (ghazi.hassen@enpc.fr)

TD & TPA :

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groupe 3 : sarah.ropert@enpc.fr

groupe 4 : alice.gribonval@enpc.fr

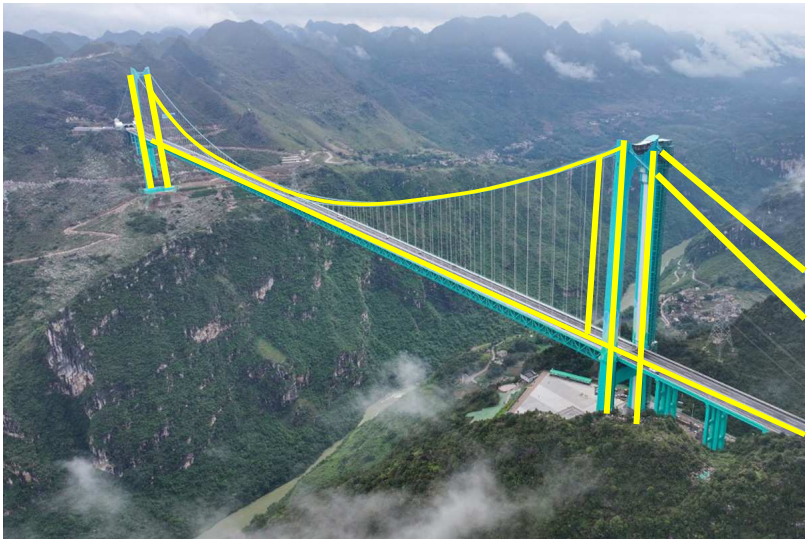
groupe 5 : samuel.guiton@enpc.fr

groupe 6 : jeremy.bleyer@enpc.fr

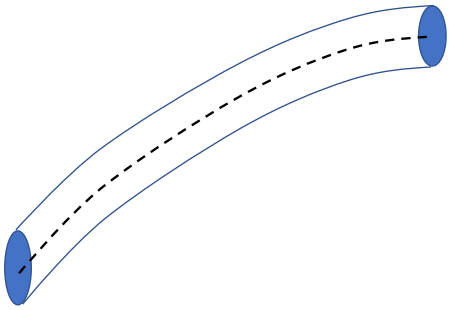
groupe 7 : suzanne.leonard@enpc.fr

groupe 8 : kianoosh.taghizadeh@enpc.fr

Modélisation géométrique

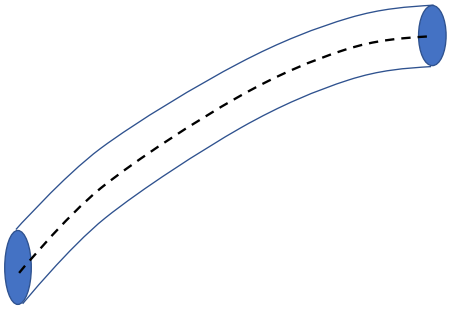


Modélisation géométrique



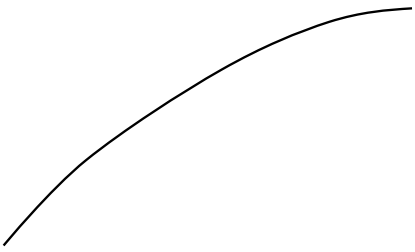
Géométrie réelle

Modélisation géométrique



modélisation

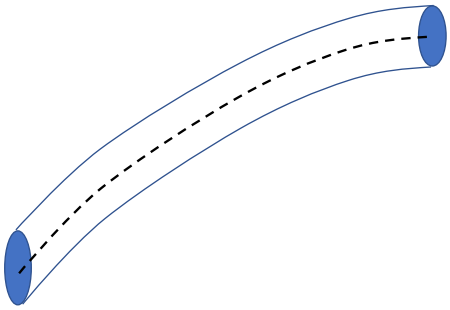
géométrique



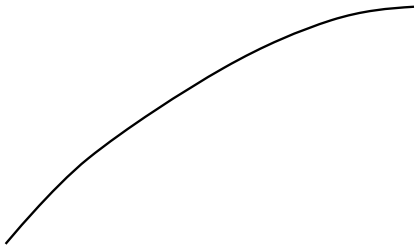
Géométrie réelle

Arc (modèle curviligne)

Modélisation géométrique



modélisation géométrique

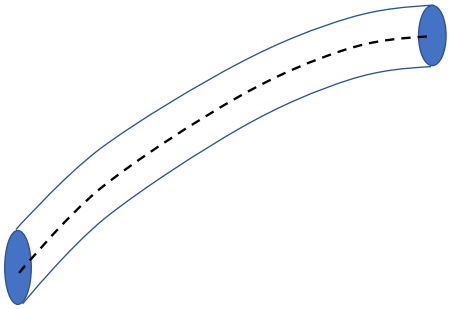


Géométrie réelle

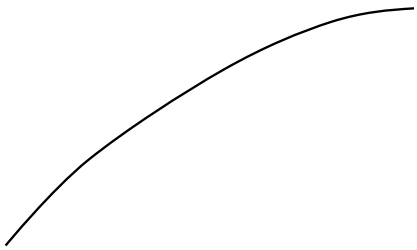
Arc (modèle curviligne)

... et la section ?

Modélisation géométrique



modélisation géométrique



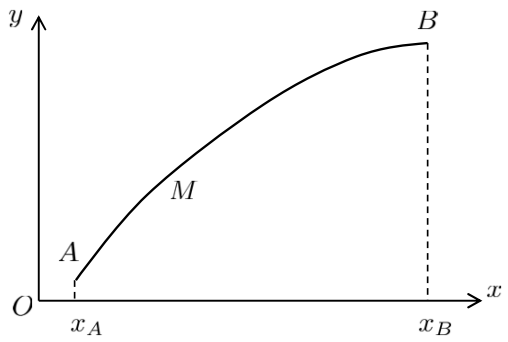
Géométrie réelle

Arc (modèle curviligne)

... et la section ? → prise en compte dans la cinématique et le comportement

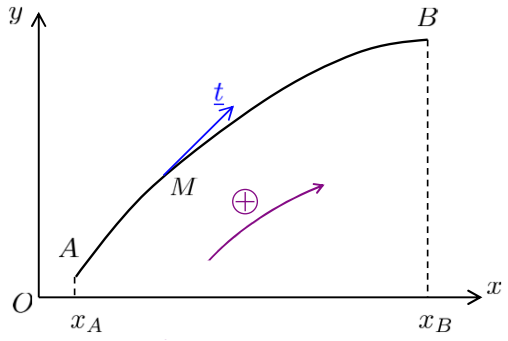
cf. cours du 24/09/26

Géométrie différentielle : cas d'un arc plan



Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

Géométrie différentielle : cas d'un arc plan



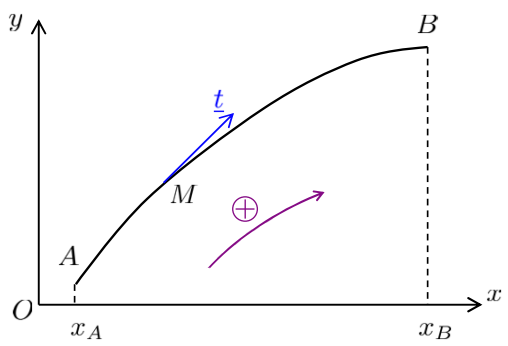
Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

autre paramétrage : abscisse curviligne

- choix arbitraire d'une origine
- choix d'un sens de parcours

Arc $AB : \{M(s_M, \underline{t}(s_M)), s_A \leq s_M \leq s_B\}$

Géométrie différentielle : cas d'un arc plan



Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

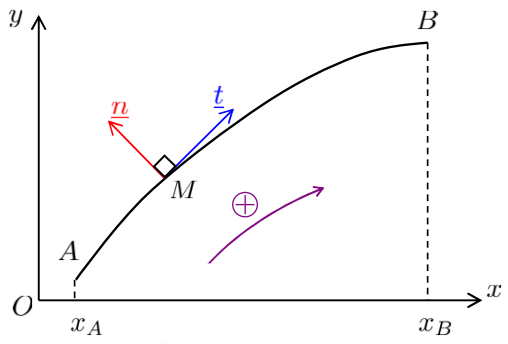
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$$\underline{t}(s_M) = \frac{\mathrm{d}\overrightarrow{OM}}{\mathrm{d}s_M}$$

Géométrie différentielle : cas d'un arc plan



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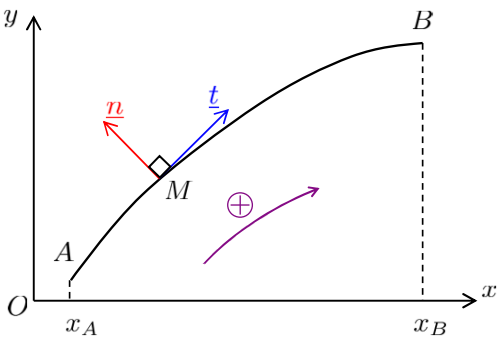
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$$\underline{t}(s_M) = \frac{\mathrm{d}\overrightarrow{OM}}{\mathrm{d}s_M} \qquad (\underline{t}, \underline{n}, \underline{e}_z) : \text{direct}$$

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Géométrie différentielle : cas d'un arc plan



Exemple 1 :

Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

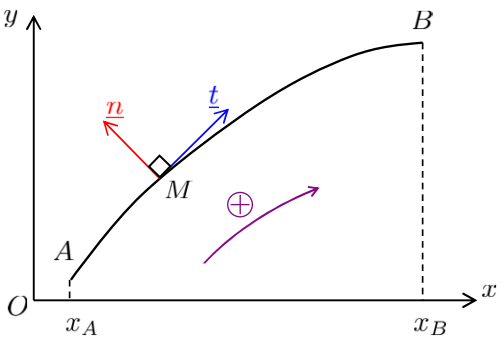
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Géométrie différentielle : cas d'un arc plan



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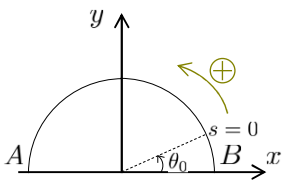
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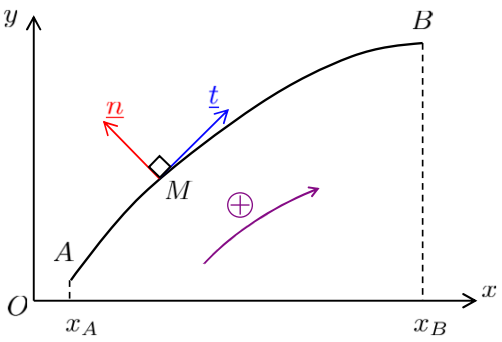
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Géométrie différentielle : cas d'un arc plan



Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

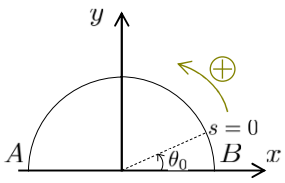
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Exemple 1 :



$s = R(\theta - \theta_0)$

Géométrie différentielle : cas d'un arc plan

Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

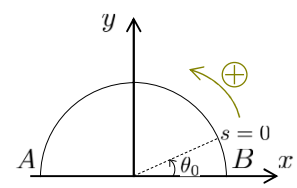
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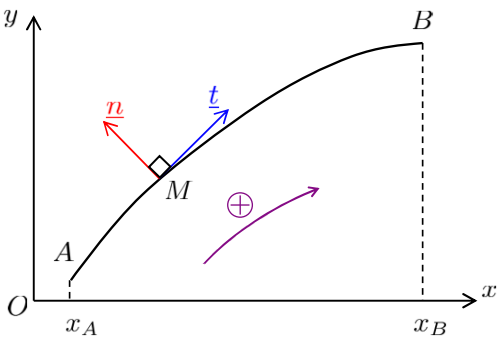
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Géométrie différentielle : cas d'un arc plan



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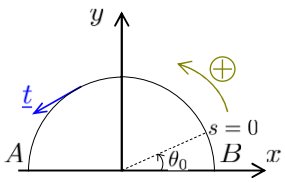
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Exemple 1 :



$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

Géométrie différentielle : cas d'un arc plan

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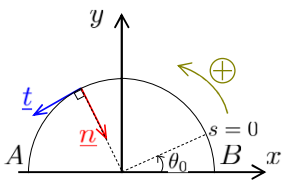
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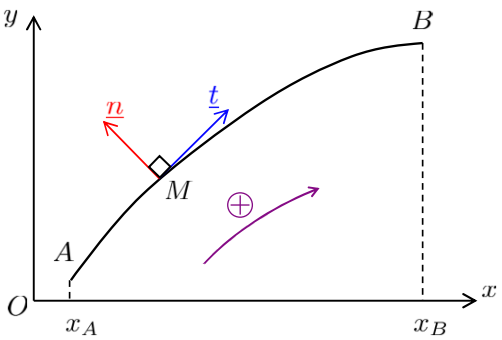
$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = -\underline{e}_r$

Géométrie différentielle : cas d'un arc plan



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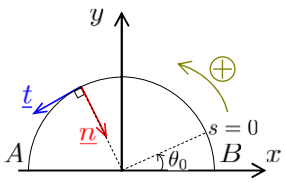
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$\underline{t}(s_M) = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s_M}$ $\qquad (\underline{t}, \underline{n}, \underline{e}_z)$: direct

Exemple 1 :



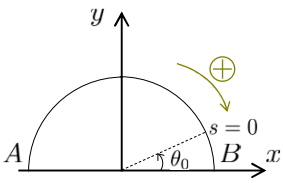
$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

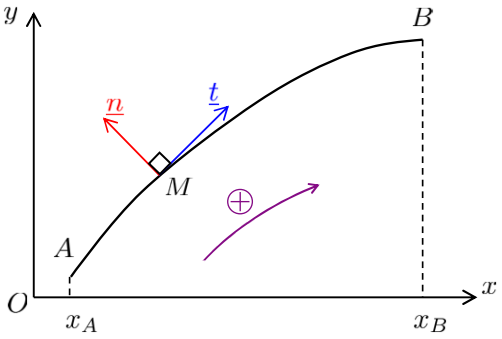
$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = -\underline{e}_r$

Exemple 2 :



Géométrie différentielle : cas d'un arc plan



Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

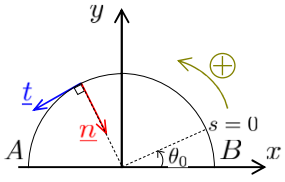
autre paramétrage : abscisse curviligne

- choix arbitraire d'une origine
- choix d'un sens de parcours

Arc $AB : \{M(s_M, \underline{t}(s_M)), s_A \leq s_M \leq s_B\}$

$\underline{t}(s_M) = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s_M}$ $\qquad (\underline{t}, \underline{n}, \underline{e}_z)$: direct

Exemple 1 :



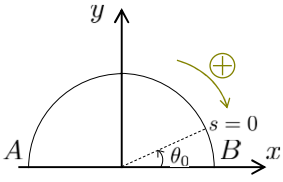
$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = -\underline{e}_r$

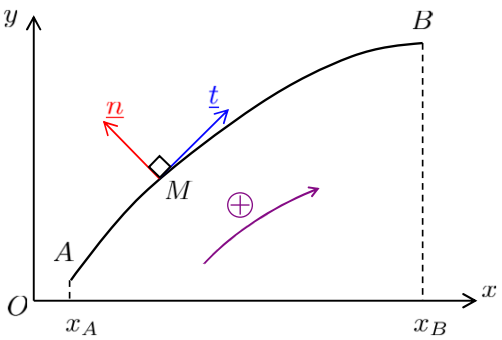
Exemple 2 :



$s = R(\theta_0 - \theta)$

$\underline{OM} = R\underline{e}_r$

Géométrie différentielle : cas d'un arc plan



Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

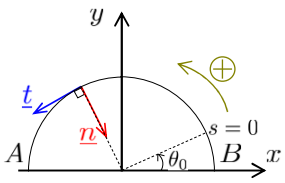
autre paramétrage : abscisse curviligne

- choix arbitraire d'une origine
- choix d'un sens de parcours

Arc $AB : \{M(s_M, \underline{t}(s_M)), s_A \leq s_M \leq s_B\}$

$\underline{t}(s_M) = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s_M}$ $\qquad (\underline{t}, \underline{n}, \underline{e}_z)$: direct

Exemple 1 :



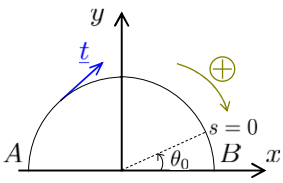
$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = -\underline{e}_r$

Exemple 2 :



$s = R(\theta_0 - \theta)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = -\underline{e}_\theta$

Géométrie différentielle : cas d'un arc plan

Arc $AB : \{M(x_M, f(x_M)), x_A \leq x_M \leq x_B\}$

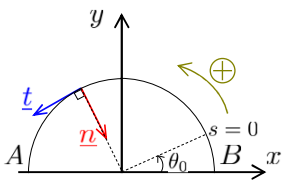
autre paramétrage : abscisse curviligne

- choix arbitraire d'une origine
- choix d'un sens de parcours

Arc $AB : \{M(s_M, \underline{t}(s_M)), s_A \leq s_M \leq s_B\}$

$\underline{t}(s_M) = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s_M}$ $\qquad (\underline{t}, \underline{n}, \underline{e}_z)$: direct

Exemple 1 :



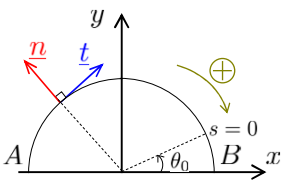
$s = R(\theta - \theta_0)$

$\underline{OM} = R\underline{e}_r$

$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = \underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = -\underline{e}_r$

Exemple 2 :



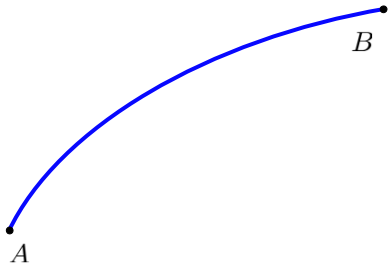
$s = R(\theta_0 - \theta)$

$\underline{OM} = R\underline{e}_r$

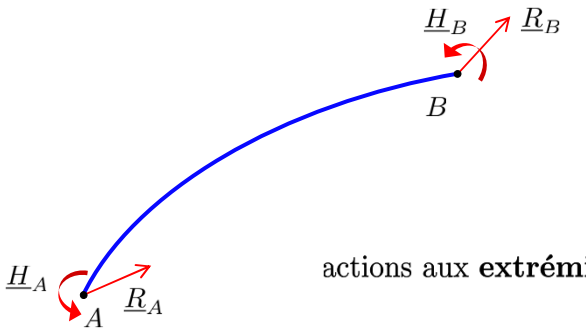
$\underline{t} = \dfrac{\mathrm{d}\underline{OM}}{\mathrm{d}s} = -\underline{e}_\theta$

$\underline{n} = \underline{e}_z \wedge \underline{t} = \underline{e}_r$

Modélisation des **efforts extérieurs** (chargement)

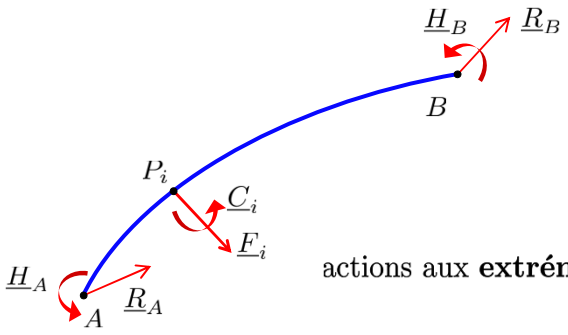


Modélisation des **efforts extérieurs** (chargement)



actions aux **extrémités** : $\begin{cases} \text{forces} : \underline{R}_A, \underline{R}_B \\ \text{moments} : \underline{H}_A, \underline{H}_B \end{cases}$

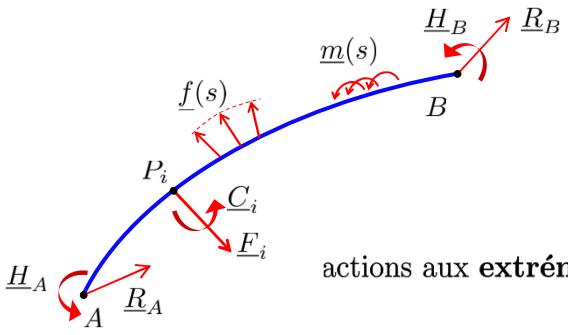
Modélisation des **efforts extérieurs** (chargement)



actions aux **extrémités** : $\begin{cases} \text{forces} : \underline{R}_A, \underline{R}_B \\ \text{moments} : \underline{H}_A, \underline{H}_B \end{cases}$

actions **concentrées** : $\begin{cases} \text{forces} : \underline{F}_i \\ \text{moments} : \underline{C}_i \end{cases}$

Modélisation des **efforts extérieurs** (chargement)

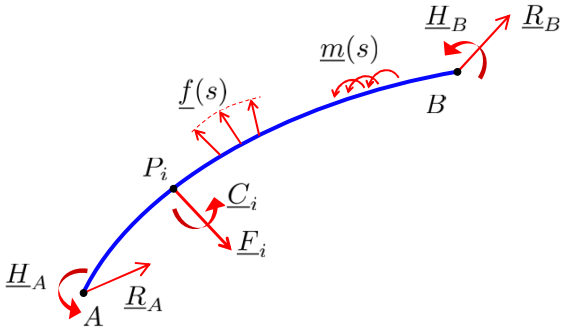


actions aux **extrémités** : $\begin{cases} \text{forces} : \underline{R}_A, \underline{R}_B \\ \text{moments} : \underline{H}_A, \underline{H}_B \end{cases}$

actions **concentrées** : $\begin{cases} \text{forces} : \underline{F}_i \\ \text{moments} : \underline{C}_i \end{cases}$

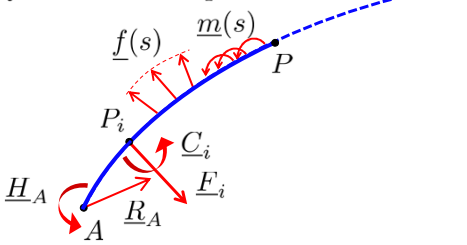
actions **réparties** : $\begin{cases} \text{densité linéique de forces} : \underline{f} \\ \text{densité linéique de moments} : \underline{m} \end{cases}$

Modélisation des **efforts intérieurs**

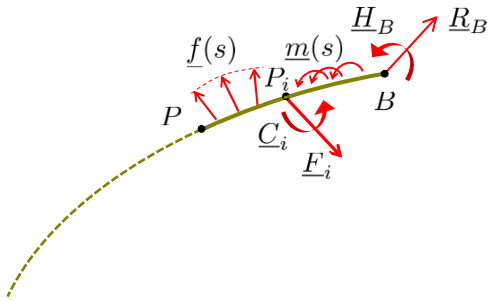


Modélisation des **efforts intérieurs**

système mécanique : arc AP

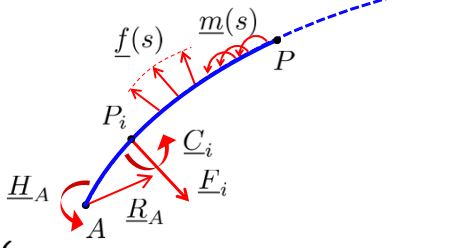


système mécanique : arc PB



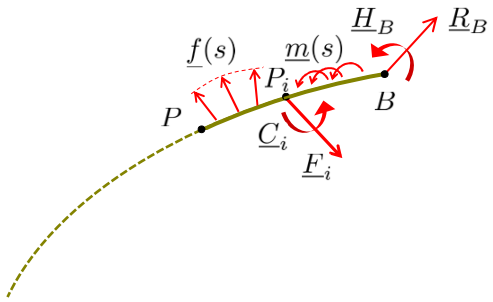
Modélisation des **efforts intérieurs**

système mécanique : arc AP



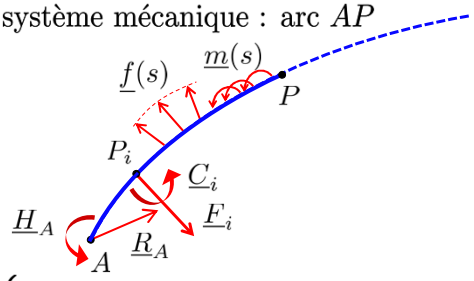
- force : $\underline{R}_A$
- moment : $\underline{H}_A$

système mécanique : arc PB



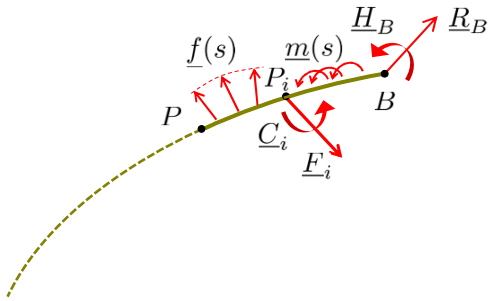
Modélisation des **efforts intérieurs**

système mécanique : arc AP



- force : $\underline{R}_A$
- moment : $\underline{H}_A$

système mécanique : arc PB

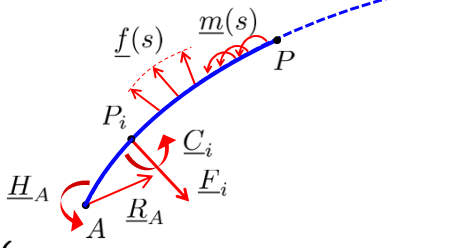


- forces : $\underline{F}_i \quad (s_A < s_i \leq s_P)$
- moments : $\underline{C}_i \quad (s_A < s_i \leq s_P)$

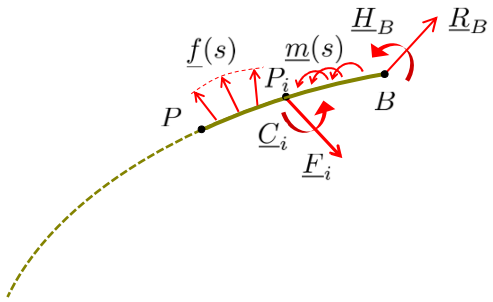
amphi1- p.11

Modélisation des **efforts intérieurs**

système mécanique : arc AP



système mécanique : arc PB



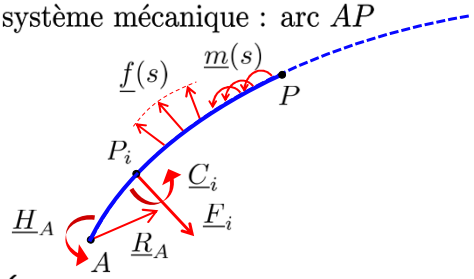
$\begin{cases} \text{force : } \underline{R}_A \\ \text{moment : } \underline{H}_A \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \quad (s_A < s_i \leq s_P) \\ \text{moments : } \underline{C}_i \quad (s_A < s_i \leq s_P) \end{cases}$

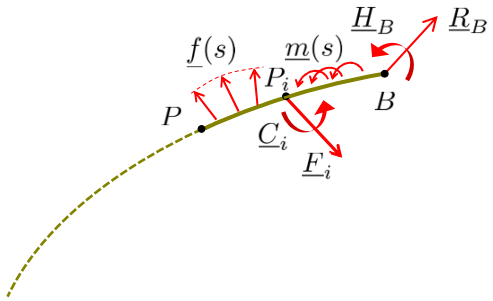
$\begin{cases} \text{forces linéiques : } \underline{f} \quad (s_A \leq s \leq s_P) \\ \text{moments linéiques : } \underline{m} \quad (s_A \leq s \leq s_P) \end{cases}$

Modélisation des **efforts intérieurs**

système mécanique : arc AP



système mécanique : arc PB



$\begin{cases} \text{force : } \underline{R}_A \\ \text{moment : } \underline{H}_A \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \quad (s_A < s_i \leq s_P) \\ \text{moments : } \underline{C}_i \quad (s_A < s_i \leq s_P) \end{cases}$

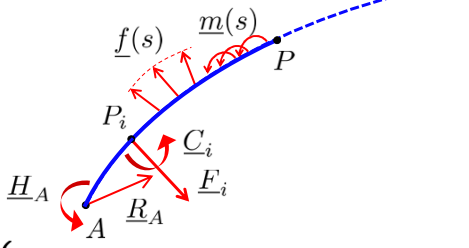
$\begin{cases} \text{forces linéiques : } \underline{f} \quad (s_A \leq s \leq s_P) \\ \text{moments linéiques : } \underline{m} \quad (s_A \leq s \leq s_P) \end{cases}$

actions de PB sur AP

amphi1- p.11

Modélisation des **efforts intérieurs**

système mécanique : arc AP



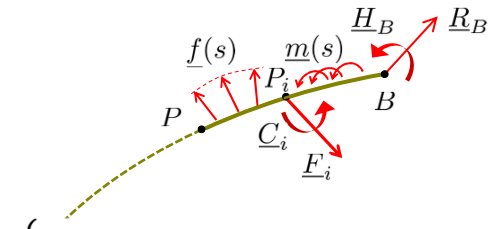
$\begin{cases} \text{force : } \underline{R}_A \\ \text{moment : } \underline{H}_A \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \ \ (s_A < s_i \leq s_P) \\ \text{moments : } \underline{C}_i \ \ (s_A < s_i \leq s_P) \end{cases}$

$\begin{cases} \text{forces linéiques : } \underline{f} \ \ (s_A \leq s \leq s_P) \\ \text{moments linéiques : } \underline{m} \ \ (s_A \leq s \leq s_P) \end{cases}$

actions de PB sur AP

système mécanique : arc PB



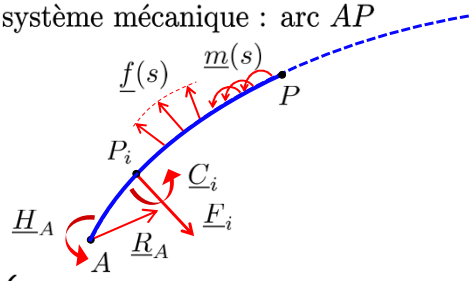
$\begin{cases} \text{force : } \underline{R}_B \\ \text{moment : } \underline{H}_B \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \ \ (s_P \leq s_i < s_B) \\ \text{moments : } \underline{C}_i \ \ (s_P \leq s_i < s_B) \end{cases}$

$\begin{cases} \text{forces linéiques : } \underline{f} \ \ (s_P \leq s \leq s_B) \\ \text{moments linéiques : } \underline{m} \ \ (s_P \leq s \leq s_B) \end{cases}$

Modélisation des **efforts intérieurs**

système mécanique : arc AP



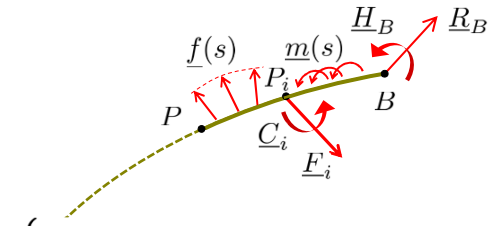
$\begin{cases} \text{force : } \underline{R}_A \\ \text{moment : } \underline{H}_A \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \ \ (s_A < s_i \leq s_P) \\ \text{moments : } \underline{C}_i \ \ (s_A < s_i \leq s_P) \end{cases}$

$\begin{cases} \text{forces linéiques : } \underline{f} \ \ (s_A \leq s \leq s_P) \\ \text{moments linéiques : } \underline{m} \ \ (s_A \leq s \leq s_P) \end{cases}$

actions de PB sur AP

système mécanique : arc PB



$\begin{cases} \text{force : } \underline{R}_B \\ \text{moment : } \underline{H}_B \end{cases}$

$\begin{cases} \text{forces : } \underline{F}_i \ \ (s_P \leq s_i < s_B) \\ \text{moments : } \underline{C}_i \ \ (s_P \leq s_i < s_B) \end{cases}$

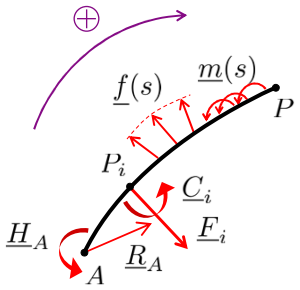
$\begin{cases} \text{forces linéiques : } \underline{f} \ \ (s_P \leq s \leq s_B) \\ \text{moments linéiques : } \underline{m} \ \ (s_P \leq s \leq s_B) \end{cases}$

actions de AP sur PB

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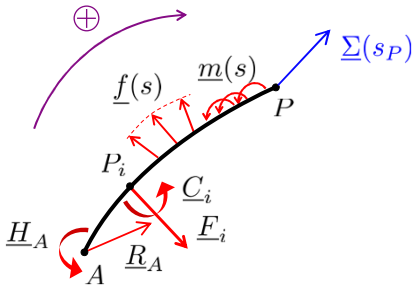
Modélisation des **efforts intérieurs**

efforts intérieurs en P : $\begin{cases} \text{résultante} : \underline{\Sigma}(s_P) \\ \text{moment} : \underline{\Gamma}(s_P) \end{cases}$



Modélisation des **efforts intérieurs**

efforts intérieurs en P : $\begin{cases} \text{résultante} : \underline{\Sigma}(s_P) \\ \text{moment} : \underline{\Gamma}(s_P) \end{cases}$

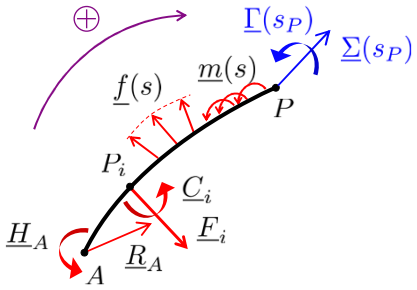


Convention :

$\underline{\Sigma}(s_P)$: **résultante** des efforts appliqués par la **partie aval** $(s \geq s_P)$ sur la **partie amont** $(s < s_P)$

Modélisation des **efforts intérieurs**

efforts intérieurs en P : $\begin{cases} \text{résultante} : \underline{\Sigma}(s_P) \\ \text{moment} : \underline{\Gamma}(s_P) \end{cases}$

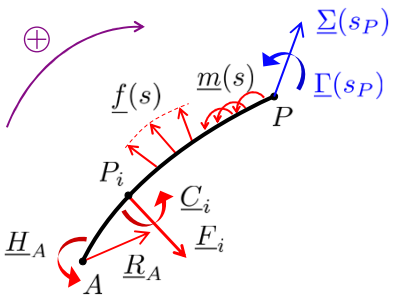


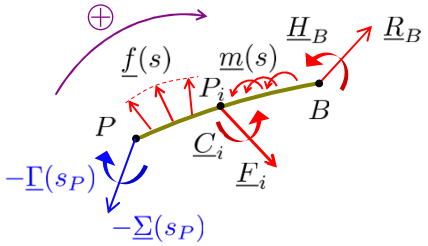
Convention :

$\underline{\Sigma}(s_P)$: **résultante** des efforts appliqués par la **partie aval** $(s \geq s_P)$ sur la **partie amont** $(s < s_P)$

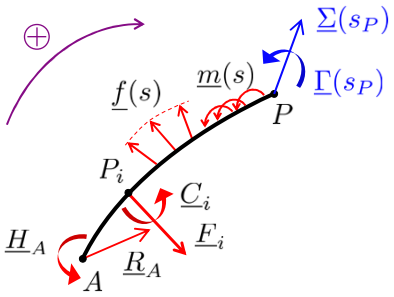
$\underline{\Gamma}(s_P)$: **moment** résultant en P des actions de la **partie aval** $(s \geq s_P)$ sur la **partie amont** $(s < s_P)$

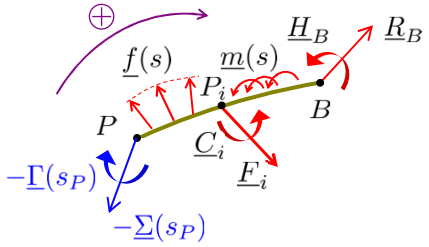
Calcul des **efforts intérieurs**





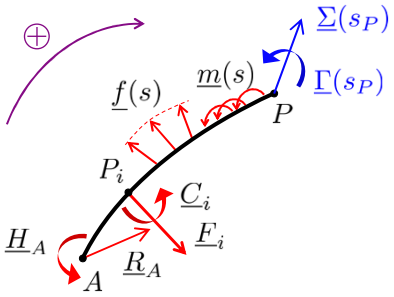
Calcul des **efforts intérieurs**

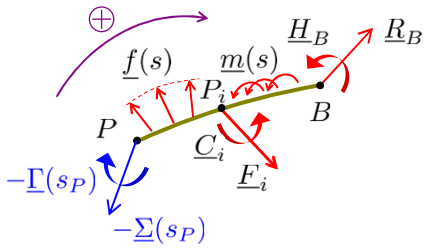




équilibre de PB (partie aval à P):

Calcul des **efforts intérieurs**

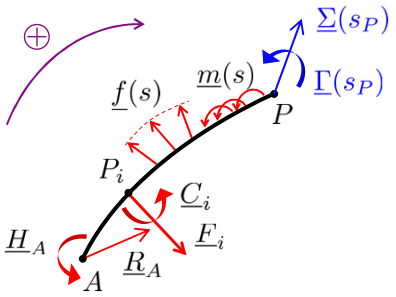


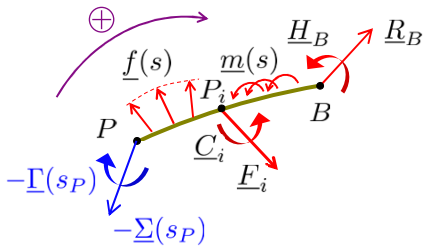


équilibre de PB (partie aval à P):

$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i > s_P} \underline{F}_i$$

Calcul des **efforts intérieurs**



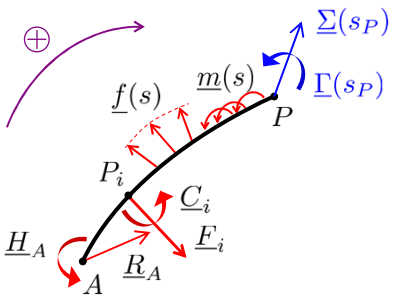


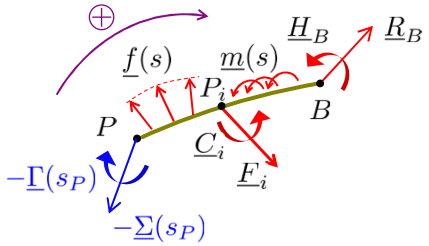
équilibre de PB (partie aval à P):

$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i > s_P} \underline{F}_i$$

$$\underline{\Gamma}(s_P) = \underline{H}_B + \underline{PB} \wedge \underline{R}_B + \int_{s_P}^{s_B} \underline{m}(s)\mathrm{d}s + \int_{s_P}^{s_B} \underline{PM} \wedge \underline{f}(s_M)\mathrm{d}s_M + \sum_{s_i > s_P} \underline{C}_i + \sum_{s_i > s_P} \underline{PP_i} \wedge \underline{F}_i$$

Calcul des **efforts intérieurs**



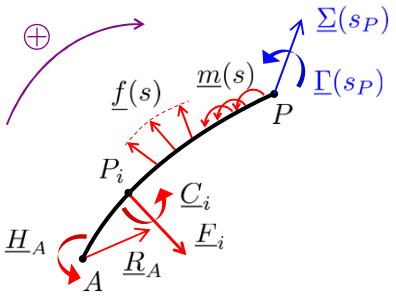


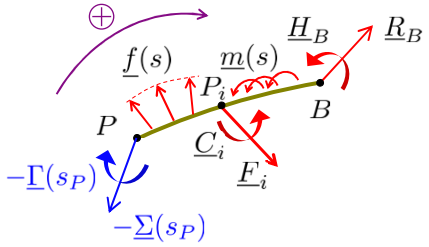
équilibre de PB (partie aval à P):

$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i > s_P} \underline{F}_i$$

$$\underline{\Gamma}(s_P) = \underline{H}_B + \underline{PB} \wedge \underline{R}_B + \int_{s_P}^{s_B} \underline{m}(s)\mathrm{d}s + \int_{s_P}^{s_B} \underline{PM} \wedge \underline{f}(s_M)\mathrm{d}s_M + \sum_{s_i > s_P} \underline{C}_i + \sum_{s_i > s_P} \underline{PP_i} \wedge \underline{F}_i$$

Calcul des **efforts intérieurs**





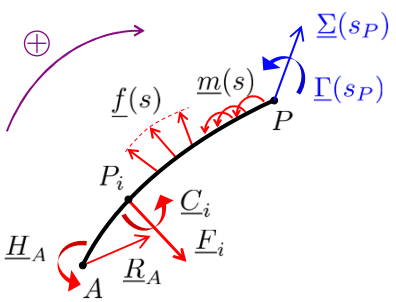
équilibre de PB (partie aval à P):

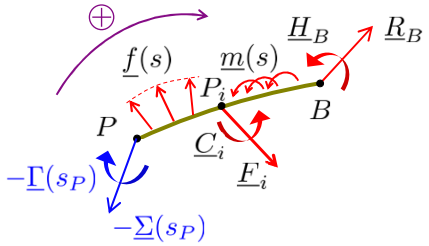
$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i>s_P} \underline{F}_i$$

$$\underline{\Gamma}(s_P) = \underline{H}_B + \underline{PB} \wedge \underline{R}_B + \int_{s_P}^{s_B} \underline{m}(s)\mathrm{d}s + \int_{s_P}^{s_B} \underline{PM} \wedge \underline{f}(s_M)\mathrm{d}s_M + \sum_{s_i>s_P} \underline{C}_i + \sum_{s_i>s_P} \underline{PP_i} \wedge \underline{F}_i$$

$$\underline{\Sigma}(s_B^-) = \underline{R}_B$$
$$\underline{\Gamma}(s_B^-) = \underline{H}_B$$

Calcul des **efforts intérieurs**





équilibre de PB (partie aval à P):

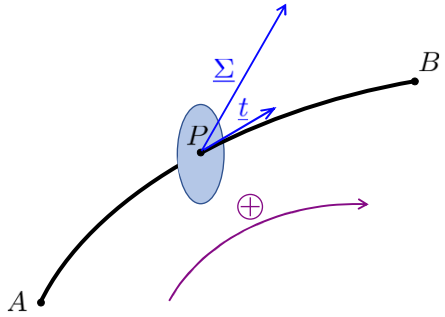
$$\underline{\Sigma}(s_P) = \underline{R}_B + \int_{s_P}^{s_B} \underline{f}(s)\mathrm{d}s + \sum_{s_i>s_P} \underline{F}_i$$

$$\underline{\Gamma}(s_P) = \underline{H}_B + \underline{PB} \wedge \underline{R}_B + \int_{s_P}^{s_B} \underline{m}(s)\mathrm{d}s + \int_{s_P}^{s_B} \underline{PM} \wedge \underline{f}(s_M)\mathrm{d}s_M + \sum_{s_i>s_P} \underline{C}_i + \sum_{s_i>s_P} \underline{PP_i} \wedge \underline{F}_i$$

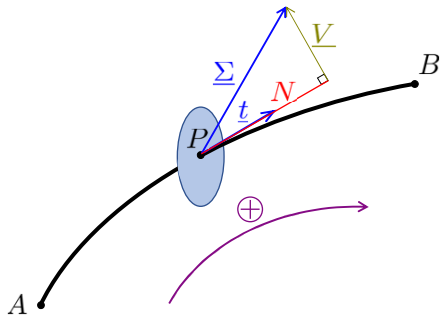
$$\underline{\Sigma}(s_B^-) = \underline{R}_B$$
$$\underline{\Gamma}(s_B^-) = \underline{H}_B$$

$$\underline{\Sigma}(s_A^+) = -\underline{R}_A$$
$$\underline{\Gamma}(s_A^+) = -\underline{H}_A$$

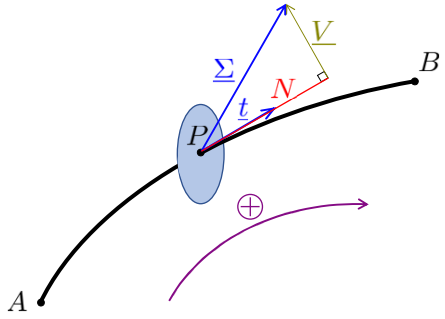
Efforts intérieurs



Efforts intérieurs



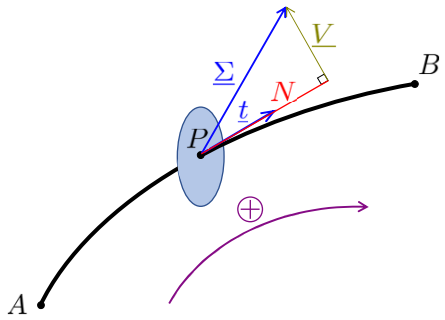
Efforts intérieurs



$\underline{\Sigma} = N\underline{t} + \underline{V}$

$\underline{V} \cdot \underline{t} = 0$

Efforts intérieurs

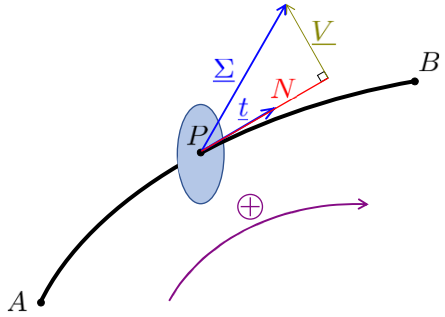


$\underline{\Sigma} = N\underline{t} + \underline{V}$

$\underline{V} \cdot \underline{t} = 0$

N : effort **normal**

Efforts intérieurs

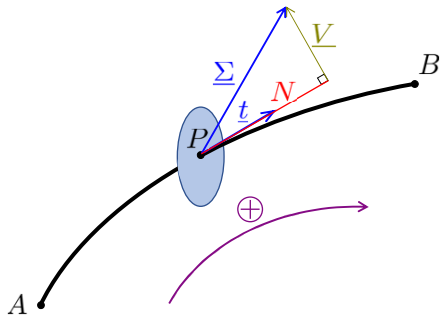


$\underline{\Sigma} = N\underline{t} + \underline{V}$

$\underline{V} \cdot \underline{t} = 0$

N : effort **normal** $\begin{cases} > 0 : \text{traction} \\ < 0 : \text{compression} \end{cases}$

Efforts intérieurs



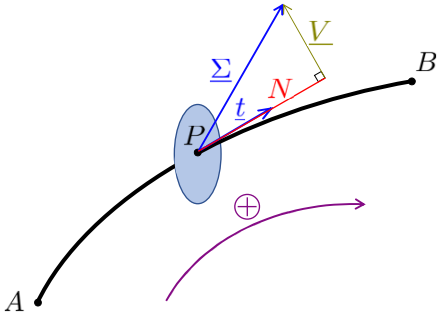
$\underline{\Sigma} = N\underline{t} + \underline{V}$

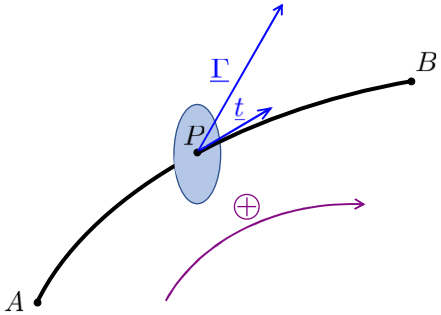
$\underline{V} \cdot \underline{t} = 0$

N : effort **normal** $\begin{cases} > 0 : \text{traction} \\ < 0 : \text{compression} \end{cases}$

$\underline{V}$: effort **tranchant**

Efforts intérieurs





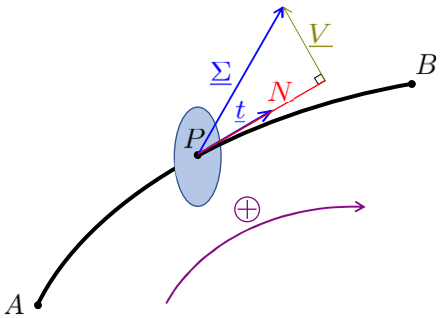
$\underline{\Sigma} = N\underline{t} + \underline{V}$

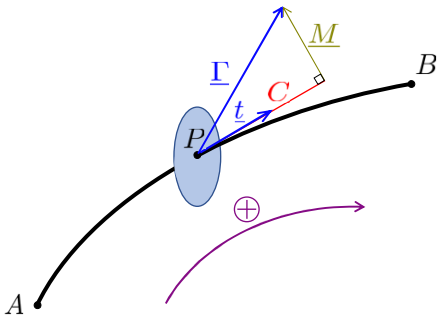
$\underline{V} \cdot \underline{t} = 0$

N : effort **normal** $\begin{cases} > 0 : \text{traction} \\ < 0 : \text{compression} \end{cases}$

$\underline{V}$: effort **tranchant**

Efforts intérieurs





$\underline{\Sigma} = N\underline{t} + \underline{V}$

$\underline{V} \cdot \underline{t} = 0$

N : effort **normal** $\begin{cases} > 0 : \text{traction} \\ < 0 : \text{compression} \end{cases}$

$\underline{V}$: effort **tranchant**

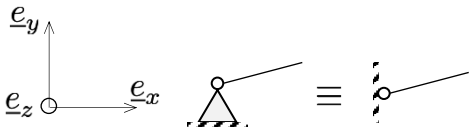
$\underline{\Gamma} = C\underline{t} + \underline{M}$

$\underline{M} \cdot \underline{t} = 0$

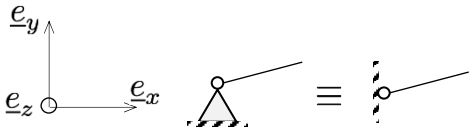
C : couple de **torsion**

$\underline{M}$: moment **fléchissant**

Conditions aux appuis : exemples

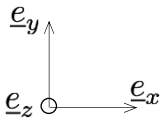


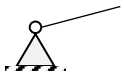
Conditions aux appuis : exemples



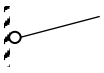
$$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$$

Conditions aux appuis : exemples

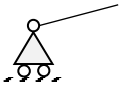




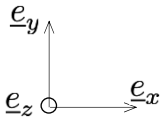
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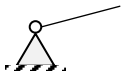


$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$

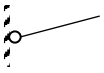


Conditions aux appuis : exemples

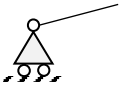




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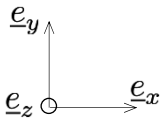


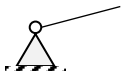
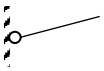
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$



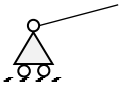
$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$

Conditions aux appuis : exemples




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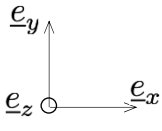
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$

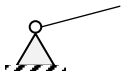
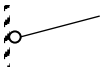


$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$

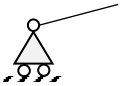


Conditions aux appuis : exemples




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$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$

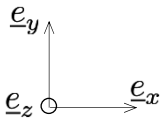


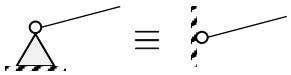
$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$



amphi1- p.15

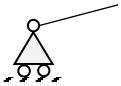
Conditions aux appuis : exemples





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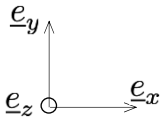
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$

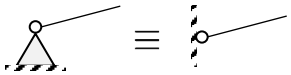


$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$



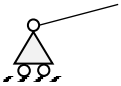
Conditions aux appuis : exemples





$\equiv$

$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$

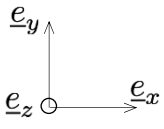


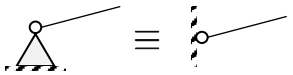
$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$



$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H} \text{ n'est pas donné} \end{cases}$

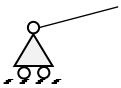
Conditions aux appuis : exemples





$\equiv$

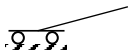
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$



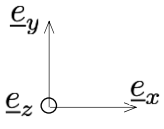
$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$

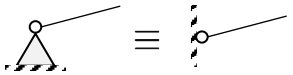


$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H} \text{ n'est pas donné} \end{cases}$



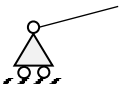
Conditions aux appuis : exemples





$\equiv$

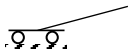
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H}^d = 0 \end{cases}$



$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H}^d = \underline{0} \end{cases}$



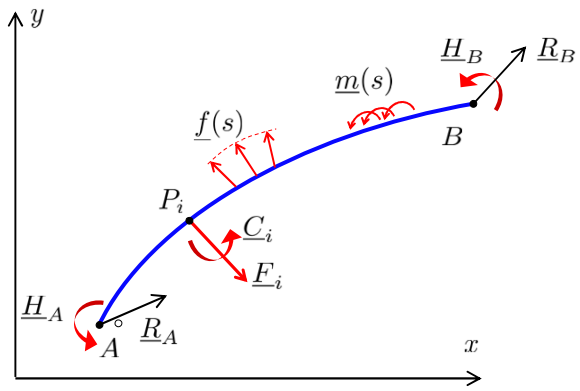
$\begin{cases} \underline{R} \text{ n'est pas donné} \\ \underline{H} \text{ n'est pas donné} \end{cases}$



$\begin{cases} R_x^d = 0 \text{ ; } R_y \text{ et } R_z \text{ ne sont pas donnés} \\ \underline{H} \text{ n'est pas donné} \end{cases}$

amphi1- p.15

Efforts intérieurs : cas d'une structure plane chargée dans son plan

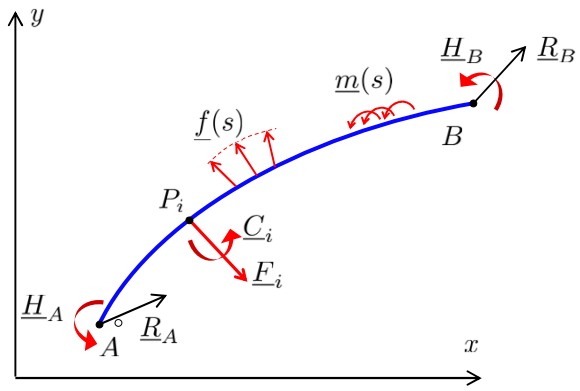


$$\underline{\Sigma} = N\underline{t} + \underline{V}$$

$$\underline{\Gamma} = C\underline{t} + \underline{M}$$

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$



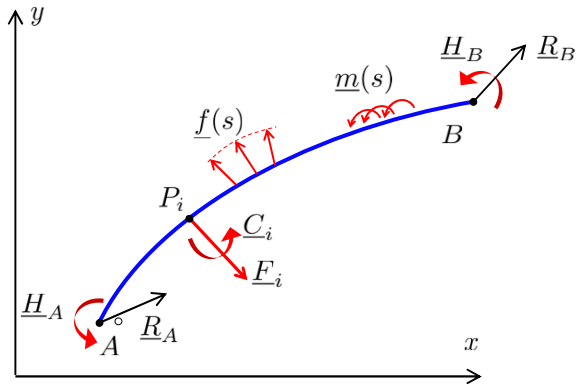
$$\underline{\Sigma} = N\underline{t} + \underline{V}$$

$$\underline{\Gamma} = C\underline{t} + \underline{M}$$

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow \begin{cases} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{cases}$



$\underline{\Sigma} = N\underline{t} + \underline{V}$

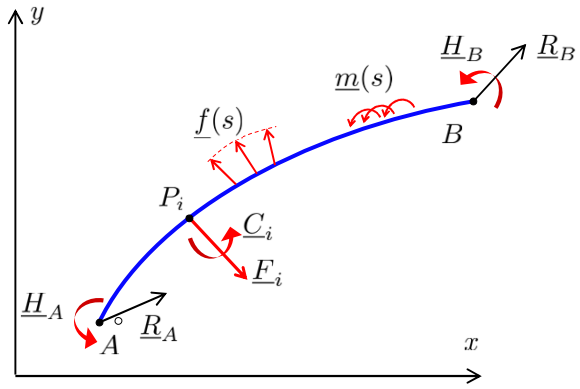
$\underline{\Gamma} = C\underline{t} + \underline{M}$

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow \begin{cases} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{cases}$

$\underline{\Sigma} \in \text{Vect}(\underline{e}_x, \underline{e}_y)$



$\underline{\Sigma} = N\underline{t} + \underline{V}$

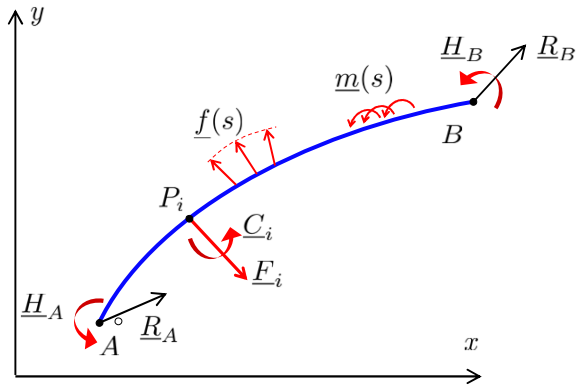
$\underline{\Gamma} = C\underline{t} + \underline{M}$

amphi1- p.16

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow$ $\left\{ \begin{array}{l} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{array} \right\}$ $\begin{array}{l} \underline{\Sigma} \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{\Gamma} \text{ porté par } \underline{e}_z \end{array}$



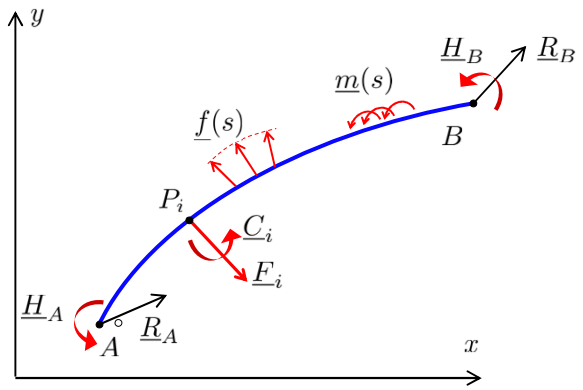
$\underline{\Sigma} = N\underline{t} + \underline{V}$

$\underline{\Gamma} = C\underline{t} + \underline{M}$

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow$ $\left\{ \begin{array}{l} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{array} \right\}$ $\begin{array}{l} \underline{\Sigma} \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{\Gamma} \text{ porté par } \underline{e}_z \end{array}$



$\underline{\Sigma} = N\underline{t} + \underline{V} = N\underline{t} + V\underline{n}$

$V = \underline{V} \cdot \underline{n}$

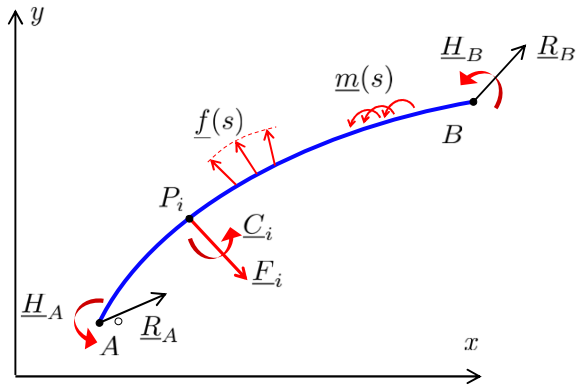
$\underline{\Gamma} = C\underline{t} + \underline{M}$

amphi1- p.16

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow$ $\left\{ \begin{array}{l} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{array} \right\}$ $\begin{array}{l} \underline{\Sigma} \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{\Gamma} \text{ porté par } \underline{e}_z \end{array}$

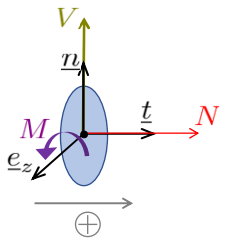


$$\underline{\Sigma} = N\underline{t} + \underline{V} = N\underline{t} + V\underline{n}$$
$$V = \underline{V} \cdot \underline{n}$$
$$\underline{\Gamma} = C\underline{t} + \underline{M} = M\underline{e}_z$$
$$M = \underline{M} \cdot \underline{e}_z$$

Efforts intérieurs : cas d'une structure plane chargée dans son plan

structure plane $\Longleftrightarrow \underline{t}(s) \in \text{Vect}(\underline{e}_x, \underline{e}_y)$

chargement plan $\Longleftrightarrow$ $\left\{ \begin{array}{l} \underline{R}_A, \underline{R}_B, \underline{f}, \underline{F}_i \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{H}_A, \underline{H}_B, \underline{m}, \underline{C}_i \text{ portés par } \underline{e}_z \end{array} \right\}$ $\begin{array}{l} \underline{\Sigma} \in \text{Vect}(\underline{e}_x, \underline{e}_y) \\ \\ \underline{\Gamma} \text{ porté par } \underline{e}_z \end{array}$

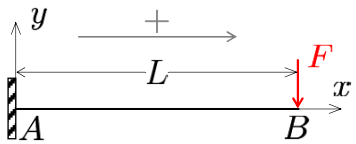


$$\underline{\Sigma} = N\underline{t} + \underline{V} = N\underline{t} + V\underline{n}$$
$$V = \underline{V} \cdot \underline{n}$$
$$\underline{\Gamma} = C\underline{t} + \underline{M} = M\underline{e}_z$$
$$M = \underline{M} \cdot \underline{e}_z$$

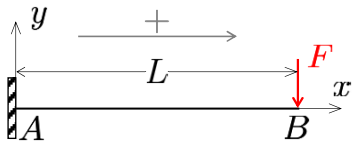
efforts intérieurs : (N, V, M)

amphi1- p.16

Exemple 1

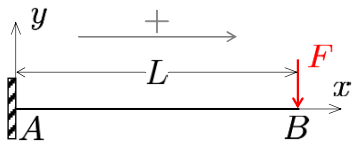


Exemple 1



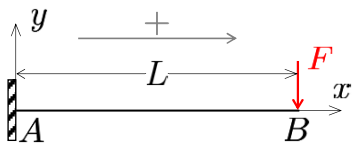
$P \in [AB]$:

Exemple 1



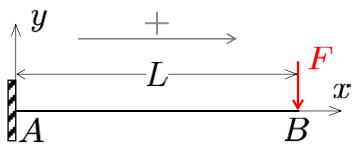
$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y \end{cases}$$

Exemple 1



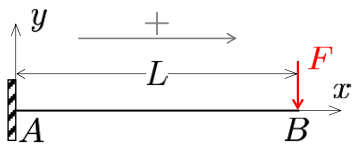
$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y \ = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y \end{cases}$$

Exemple 1



$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y \ = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y \ = F(x_P - L)\underline{e}_z \end{cases}$$

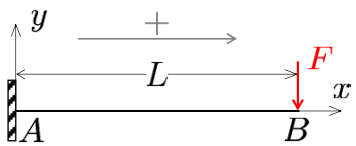
Exemple 1



$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y \ = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y \ = F(x_P - L)\underline{e}_z \end{cases}$$

$$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y$$

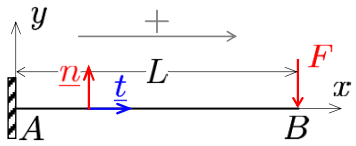
Exemple 1



$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y = F(x_P - L)\underline{e}_z \end{cases}$$

$$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = FL\underline{e}_z$$

Exemple 1

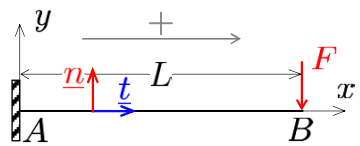


$$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$$

$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y = F(x_P - L)\underline{e}_z \end{cases}$$

$$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = FL\underline{e}_z$$

Exemple 1



$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

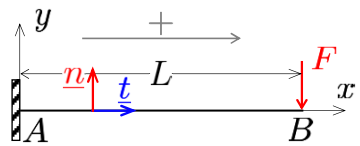
$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y = F(x_P - L)\underline{e}_z \end{cases}$$

$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y$ $\qquad$ $\underline{H}_A = -\underline{\Gamma}(A) = FL\underline{e}_z$



$N(x) = 0$

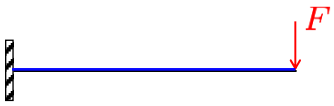
Exemple 1



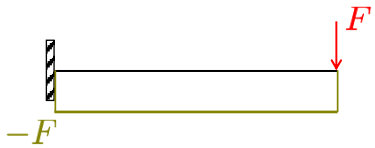
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y = F(x_P - L)\underline{e}_z \end{cases}$$

$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y$ $\qquad$ $\underline{H}_A = -\underline{\Gamma}(A) = FL\underline{e}_z$

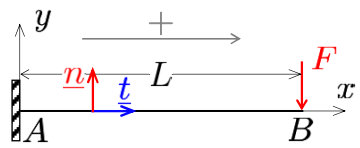


$N(x) = 0$



$V(x) = -F$

Exemple 1



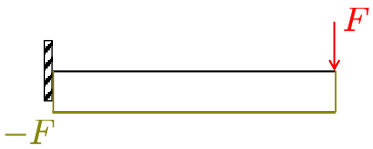
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in [AB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y \ = (L - x_P)\underline{e}_x \wedge -F\underline{e}_y \ = F(x_P - L)\underline{e}_z \end{cases}$$

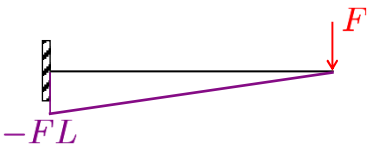
$\underline{R}_A = -\underline{\Sigma}(A) = F\underline{e}_y$ $\qquad$ $\underline{H}_A = -\underline{\Gamma}(A) = FL\underline{e}_z$



$N(x) = 0$

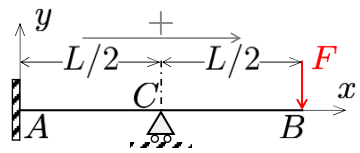


$V(x) = -F$



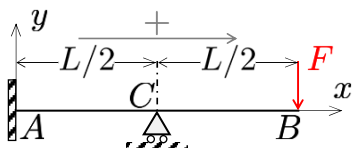
$M(x) = F(x - L)$

Exemple 2



$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

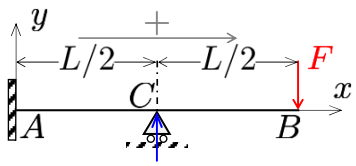
Exemple 2



$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

Exemple 2

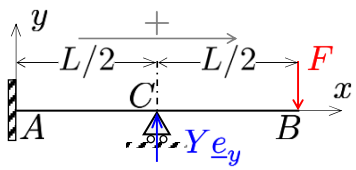


$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

Exemple 2

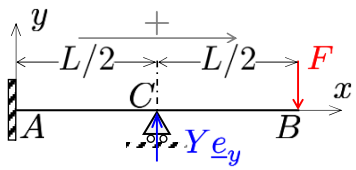


$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

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Exemple 2



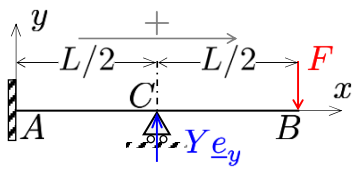
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y$

Exemple 2



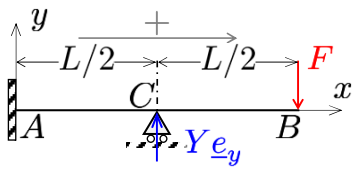
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = (F(x_P - L) + Y(L/2 - x_P))\,\underline{e}_z$

Exemple 2



$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

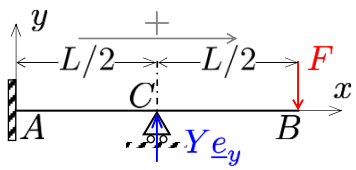
$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

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$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = (F(x_P - L) + Y(L/2 - x_P))\,\underline{e}_z$

$\underline{R}_A = -\underline{\Sigma}(A) = (F - Y)\underline{e}_y$

Exemple 2



$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

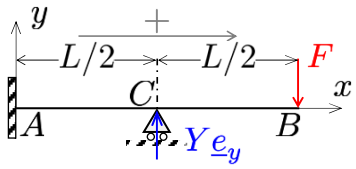
$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = (F(x_P - L) + Y(L/2 - x_P))\ \underline{e}_z$

$\underline{R}_A = -\underline{\Sigma}(A) = (F - Y)\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = (F - Y/2)L\underline{e}_z$

Exemple 2



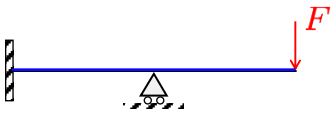
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

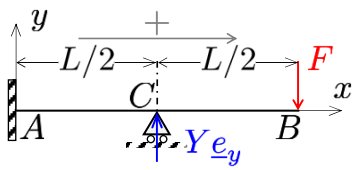
$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = (F(x_P - L) + Y(L/2 - x_P))\ \underline{e}_z$

$\underline{R}_A = -\underline{\Sigma}(A) = (F - Y)\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = (F - Y/2)L\underline{e}_z$



$N(x) = 0$

Exemple 2



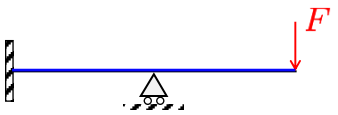
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

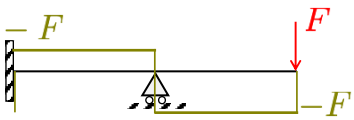
$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = F(x_P - L)\ \underline{e}_z \end{cases}$$

$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = (F(x_P - L) + Y(L/2 - x_P))\,\underline{e}_z$

$\underline{R}_A = -\underline{\Sigma}(A) = (F - Y)\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = (F - Y/2)L\underline{e}_z$

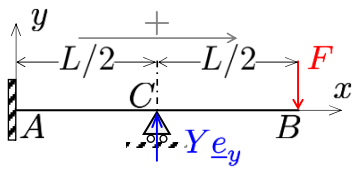




$N(x) = 0$

$$V(x) = \begin{cases} Y - F \text{ sur } [AC[\\ -F \text{ sur }]CB] \end{cases}$$

Exemple 2



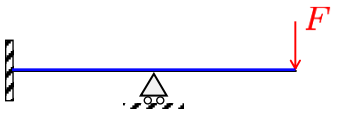
$\underline{t} = \underline{e}_x \ ; \ \underline{n} = \underline{e}_y$

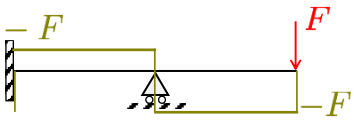
$$P \in]CB] : \begin{cases} \underline{\Sigma}(P) = -F\underline{e}_y \\ \underline{\Gamma}(P) = \boxed{F(x_P - L)}\ \underline{e}_z \end{cases}$$

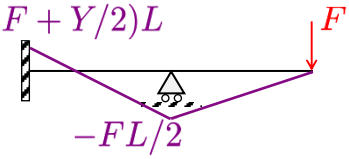
$P \in [AC[: \underline{\Sigma}(P) = (Y - F)\underline{e}_y$

$P \in [AC] : \underline{\Gamma}(P) = \underline{PB} \wedge -F\underline{e}_y + \underline{PC} \wedge Y\underline{e}_y \quad = \boxed{(F(x_P - L) + Y(L/2 - x_P))}\,\underline{e}_z$

$\underline{R}_A = -\underline{\Sigma}(A) = (F - Y)\underline{e}_y \qquad \underline{H}_A = -\underline{\Gamma}(A) = (F - Y/2)L\underline{e}_z$







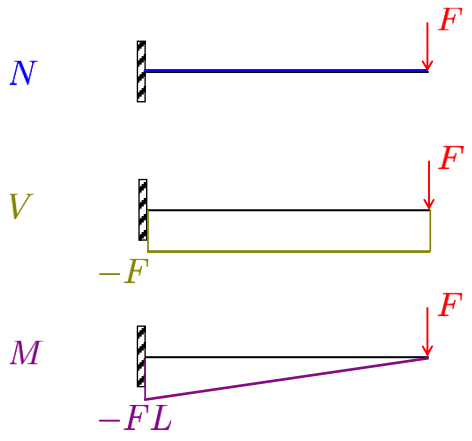
$N(x) = 0$

$$V(x) = \begin{cases} Y - F \text{ sur } [AC[\\ -F \text{ sur }]CB] \end{cases}$$

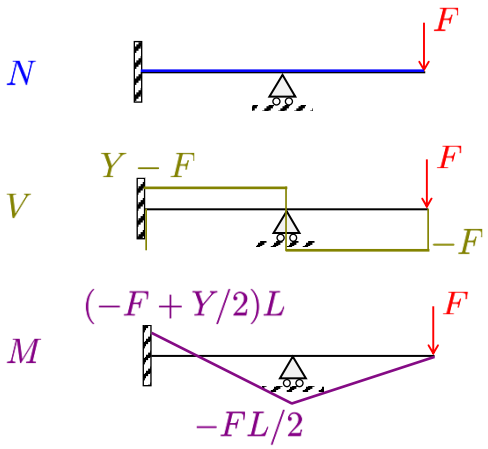
$$M(x_P) = \begin{cases} [AC] : \\ [CB] : \end{cases}$$

Commentaires

Exemple 1

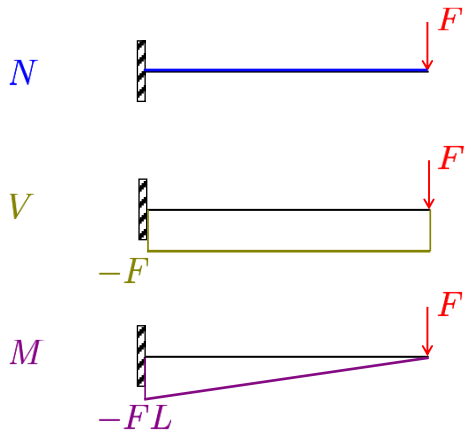


Exemple 2



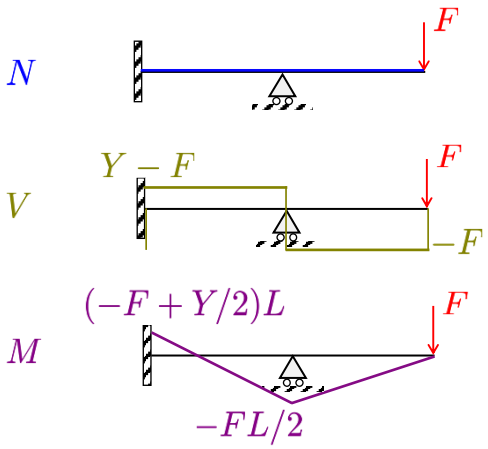
Commentaires

Exemple 1



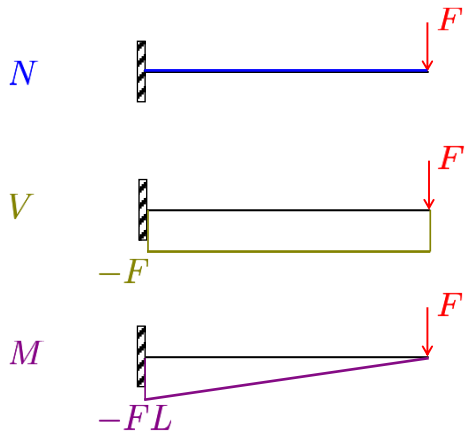
isostatique

Exemple 2



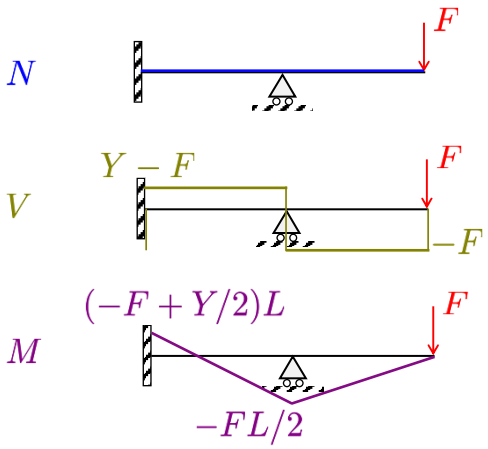
Commentaires

Exemple 1



isostatique

Exemple 2



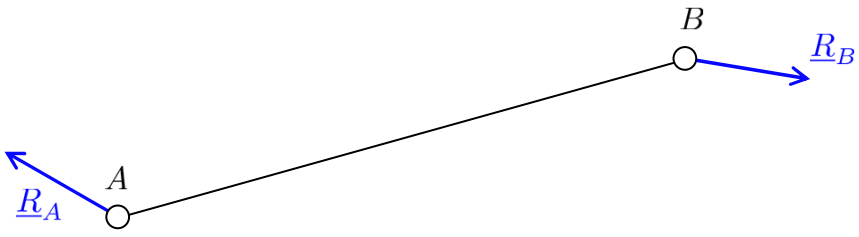
hyperstatique de degré 1

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités

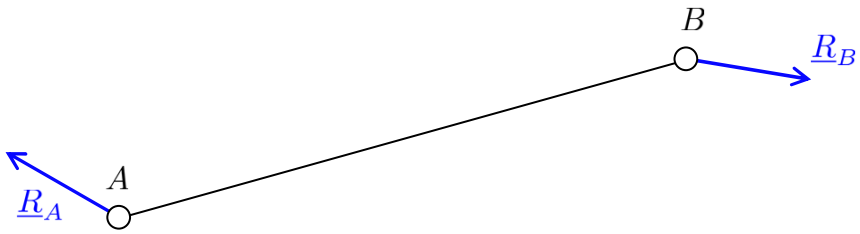


Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



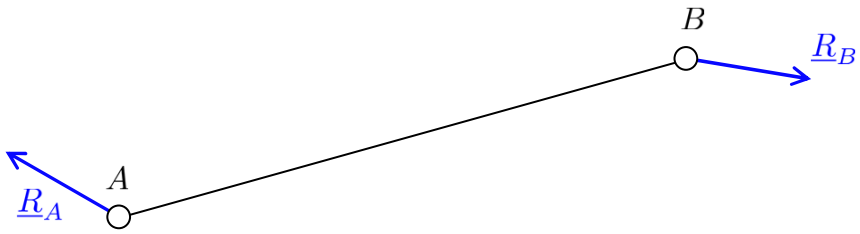
$$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0}$$

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



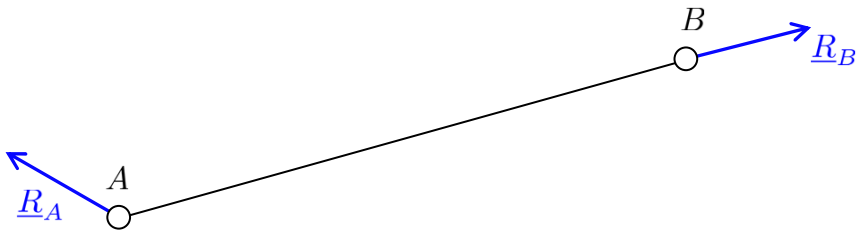
$$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0} \quad \Longrightarrow \underline{R}_B \,//\, \underline{t}$$

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



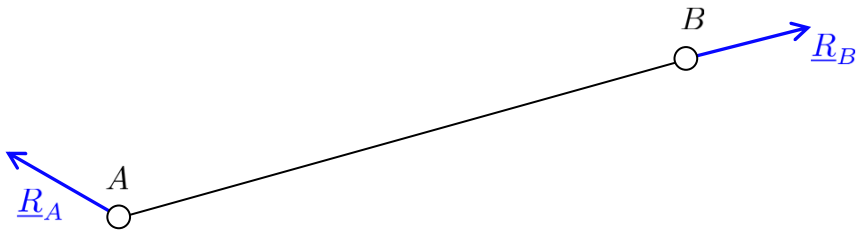
$$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0} \quad \Longrightarrow \underline{R}_B \,//\, \underline{t}$$

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



$$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0} \quad \Longrightarrow \underline{R}_B \,//\, \underline{t}$$

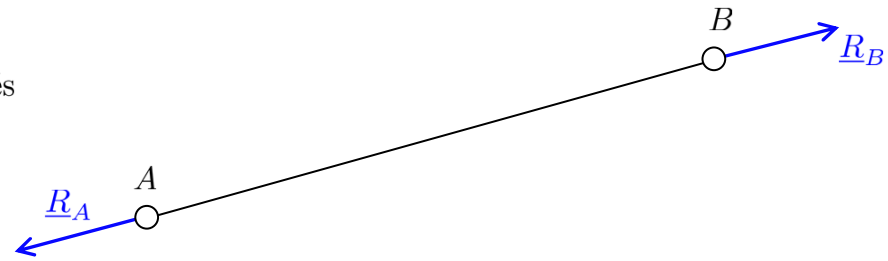
$$\underline{R}_A + \underline{R}_B = \underline{0}$$

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0} \quad \Longrightarrow \underline{R}_B \,//\, \underline{t}$

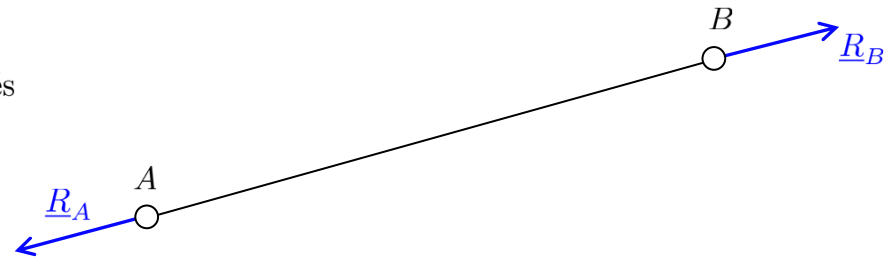
$\underline{R}_A + \underline{R}_B = \underline{0}$

Compléments : Cas des structures réticulées

arc rectiligne

+

chargement aux extrémités



$\underline{H}_A = -\underline{\Gamma}(s_A) = -\underline{AB} \wedge \underline{R}_B = \underline{0} \quad \Longrightarrow \underline{R}_B \,//\, \underline{t}$

$\underline{R}_A + \underline{R}_B = \underline{0}$

$V = 0 \;\; ; \;\; M = 0$